

The Oil Lamp Guide

Five ways to make light from what's already in your kitchen

The reel showed one jar, one wick, and a flame that outlasted the fear. Here are five complete lamp designs — from the simplest two-minute build to the most refined — using only what you already have.

WHICH OILS WORK AND HOW LONG THEY BURN

- **Olive oil (extra virgin or refined)** — burns cleanly with minimal smoke. ~4–6 hours per 1/3 cup. The traditional lamp oil of the Mediterranean for 3,000 years.
- **Vegetable oil (canola, sunflower)** — slightly more smoke than olive. ~3–5 hours per 1/3 cup. Widely available and affordable.
- **Coconut oil** — must be melted first (liquid above 24 °C/76 °F). Burns cleanly. ~4–5 hours per 1/3 cup. Solidifies in cold rooms; set the lamp near warmth.
- **Lard or rendered animal fat** — burns well when fully liquid. More smoke than plant oils. Traditional use in colder climates. ~3–4 hours per 1/3 cup.
- **Butter** — not recommended. Burns unevenly, produces significant smoke, and spatters. Use only as a last resort.
- **Never use** — gasoline, kerosene, rubbing alcohol, or acetone in these open-wick designs. These are not lamp oils; they will flare and cause fires.

WICK SOURCES FROM YOUR HOME

- **Cotton balls** — twisted tightly into a rope ~1 cm diameter. The most available material.
 - **Old cotton T-shirt or shirt** — tear a strip 1–2 cm wide from a seam. 100% cotton only; synthetics melt and produce toxic smoke.
 - **Cotton shoelace** — round shoelaces work well. Remove any plastic tip.
 - **Twisted cotton twine** — natural twine from a hardware store or kitchen drawer.
 - **Avoid** — polyester, nylon, or blended fabrics. They melt, smell chemical, and can drip flaming material.
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DESIGN 1 — THE JAR LAMP (2-MINUTE BUILD)

- Any glass jar with a wide mouth (mason jar, pasta sauce jar, jam jar).
 - Fill the jar 1/3 full with cooking oil.
 - Twist a cotton strip into a wick 15–20 cm long.
 - Drop one end into the oil. Drape the other end over the rim, or hold it upright with a small piece of foil folded over the rim.
 - Leave 1 cm of wick above the oil level. Let it soak for 2 minutes before lighting.
 - **Burn time** — 4–6 hours per 1/3 cup of olive oil. Top up oil without extinguishing.
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DESIGN 2 — THE SHALLOW DISH LAMP

- **Any shallow ceramic or metal dish** — a small plate, a can lid, or a wide bowl.
 - Pour 1 cm depth of oil into the dish.
 - Twist the wick and lay it flat across the dish so one end trails into the oil and the other end rests on the rim as the flame point.
 - A small stone or metal nut placed at the center can hold the wick submerged while the tip remains raised.
 - **Lower profile than the jar lamp** — more stable and harder to tip. Good for placement on the floor.
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DESIGN 3 — THE TIN CAN LAMP

- Use an empty tin can (tuna, tomato paste) with no sharp edges.
 - Half-fill with oil.
 - **Punch a small hole near the top rim with a nail** — just large enough to thread the wick through.
 - Feed the wick through so 1 cm protrudes outside and the rest hangs inside the oil.
 - **The can walls shield the flame from wind** — the most wind-resistant of the five designs. Good for outdoor use.
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DESIGN 4 — THE BOTTLE FLOAT LAMP

- A glass bottle (wine, oil, or sauce bottle) with a narrow neck.
 - Fill 1/3 with oil.
 - Cut a square of foil 5 × 5 cm. Poke a small hole in the center.
 - Feed the wick through the foil hole. The foil rests across the bottle neck, holding the wick upright with 1 cm above the foil.
 - The narrow neck limits oxygen to the oil surface, making this the safest design for indoor use near curtains or papers.
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DESIGN 5 — THE CLAY PINCH-POT LAMP

- **If you have air-dry clay or modelling clay, pinch a small cup with a spout** — the traditional Roman lucerna shape.
 - **Allow to dry fully (24–48 hours for air-dry clay). Unfired clay can crack when filled with warm oil** — test with water first.
 - Fill with oil and use a short twisted wick resting in the spout.
 - **The oldest lamp design in human history** — evidence from 10,000 BCE. If clay is not available, skip this design.
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WICK CARE AND TRIMMING

- **Trim the wick to 5–8 mm (1/4 inch) above the oil before each use** — a long wick produces excessive smoke and a large, unstable flame.
 - **If the flame smokes** — wick is too long. Pinch or trim.
 - **If the flame is too small or goes out** — wick has fallen below the oil level. Pull it up gently with tweezers when the flame is extinguished.
 - **Wick end chars over time** — trim the charred tip before relighting to restore a clean flame.
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SAFE EXTINGUISHING AND STORAGE

- Extinguish by placing a small metal lid or piece of foil over the flame to starve it of oxygen. Blowing out olive oil flames can scatter hot droplets.
 - Allow the lamp to cool completely before moving or refilling.
 - Store oil in a sealed container away from heat and light.
 - Never leave a lamp unattended or burning while sleeping.
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KEEP THIS IN MIND

Pre-make two jar lamps and store them with oil already in and wicks pre-soaked. They light in 10 seconds. The man who has already prepared does not fumble in the dark.

Comment GINGER tomorrow — Monday's guide covers three plants that have been treating common ailments since long before the pharmacy existed.

SOURCES Smithsonian Magazine — History of the Oil Lamp · U.S. Fire Administration — Candle and open-flame safety guidelines (adapted for oil wicks) · Traditional lamp oil burn-time data based on standard calorimetry values for triglyceride-based fats